

CSE 400: Fundamentals of Probability in Computing

Lecture 20: Linear Transformations and Expectations of Multiple Random Variables

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Overview

This lecture focuses on the statistical representation of multiple random variables, their transformations, and multivariate Gaussian distributions. The key objective is to understand how random vectors behave under linear transformations and how their mean and covariance evolve.

Key Topics Covered:

- Random vectors
- Mean vector
- Correlation and covariance matrices
- Linear transformation: $Y = AX + b$
- Transformation of mean and covariance
- Multivariate Gaussian distributions
- Case study: Smart City system

1 Random Vector Representation

1.1 Definition

For multiple random variables $X_1, X_2, \dots, X_N$, we define a random vector:

$$\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{bmatrix}$$

- A **random vector** is a compact representation of multiple random variables.
- It allows vector/matrix-based analysis.

Summary: Converts multiple RVs into a structured mathematical object.

2 Mean Vector

2.1 Definition

$$\boldsymbol{\mu}_X = \mathbb{E}[\mathbf{X}]$$

$$\boldsymbol{\mu}_X = \begin{bmatrix} \mathbb{E}[X_1] \\ \mathbb{E}[X_2] \\ \vdots \\ \mathbb{E}[X_N] \end{bmatrix}$$

- Represents average behavior of each component.

Key Insight: Mean vector gives the center of the distribution.

3 Correlation Matrix

3.1 Definition

$$R_{XX} = \mathbb{E}[\mathbf{X}\mathbf{X}^T]$$

$$R_{XX}(i, j) = \mathbb{E}[X_i X_j]$$

$$R_{XX} = \begin{bmatrix} \mathbb{E}[X_1 X_1] & \mathbb{E}[X_1 X_2] & \cdots \\ \mathbb{E}[X_2 X_1] & \mathbb{E}[X_2 X_2] & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

- Symmetric matrix

Summary: Measures joint behavior including mean effects.

4 Covariance Matrix

4.1 Definition

$$C_{XX} = \mathbb{E}[(\mathbf{X} - \boldsymbol{\mu}_X)(\mathbf{X} - \boldsymbol{\mu}_X)^T]$$

$$C_{XX} = \begin{bmatrix} \text{Var}(X_1) & \text{Cov}(X_1, X_2) \\ \text{Cov}(X_2, X_1) & \text{Var}(X_2) \end{bmatrix}$$

- Diagonal = variances
- Off-diagonal = covariances

Key Property:

$$\text{Cov}(X_i, X_j) = \text{Cov}(X_j, X_i)$$

Summary: Captures dependencies between variables.

5 Linear Transformation

5.1 Model

$$\mathbf{Y} = A\mathbf{X} + \mathbf{b}$$

- A = transformation matrix
- $\mathbf{b}$ = shift vector

Interpretation:

- A scales/rotates
- b shifts

6 Mean Transformation

6.1 Derivation

$$\boldsymbol{\mu}_Y = \mathbb{E}[Y] = \mathbb{E}[AX + b]$$

$$= A\mathbb{E}[X] + b$$

$$\boxed{\boldsymbol{\mu}_Y = A\boldsymbol{\mu}_X + b}$$

Summary: Mean transforms linearly.

7 Correlation Transformation

$$\begin{aligned} R_{YY} &= \mathbb{E}[YY^T] \\ &= \mathbb{E}[(AX + b)(AX + b)^T] \\ &= AR_{XX}A^T + A\mu_X b^T + b\mu_X^T A^T + bb^T \end{aligned}$$

Key Result:

$$R_{YY} = AR_{XX}A^T + A\mu_X b^T + b\mu_X^T A^T + bb^T$$

8 Covariance Transformation

8.1 Key Observation

$$Y - \mu_Y = A(X - \mu_X)$$

8.2 Derivation

$$\begin{aligned} C_{YY} &= \mathbb{E}[(Y - \mu_Y)(Y - \mu_Y)^T] \\ &= AC_{XX}A^T \end{aligned}$$

$$C_{YY} = AC_{XX}A^T$$

Important Note: Shift b does not affect covariance.

9 Summary of Linear Transformation

If $Y = AX + b$:

- Mean: $\mu_Y = A\mu_X + b$
- Covariance: $C_{YY} = AC_{XX}A^T$
- Correlation: includes additional terms with b

10 Numerical Example

Given:

$$\mu_X = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}, \quad A = \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 1 \\ -1 & 1 & 0 \end{bmatrix}$$

10.1 Mean

$$\boldsymbol{\mu}_Y = A\boldsymbol{\mu}_X = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}$$

10.2 Covariance

$$C_{YY} = R_{YY} - \boldsymbol{\mu}_Y \boldsymbol{\mu}_Y^T$$

$$C_{YY} = \begin{bmatrix} 16 & -8 & -8 \\ -8 & 10 & -2 \\ -8 & -2 & 10 \end{bmatrix}$$

Summary: Practice matrix multiplication carefully.

11 Gaussian Distributions

11.1 1D Gaussian

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

11.2 2D Gaussian

$$f_{XY}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp(\dots)$$

12 Multivariate Gaussian

12.1 Definition

$$\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, C)$$

$$f_X(x) = \frac{1}{\sqrt{(2\pi)^N \det(C)}} \exp\left(-\frac{1}{2}(x-\boldsymbol{\mu})^T C^{-1}(x-\boldsymbol{\mu})\right)$$

Components:

- $\boldsymbol{\mu}$: mean vector
- C : covariance matrix

Summary: Fully described by mean + covariance.

13 Special Case: Uncorrelated Gaussian

If:

$$\text{Cov}(X_i, X_j) = 0 \quad i \neq j$$

Then:

$$f_X(x) = \prod_{i=1}^N \text{Gaussian}(x_i)$$

Insight: Covariance diagonal $\Rightarrow$ independent structure.

14 Homework Result

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$$

15 Case Study: Smart City System

15.1 Random Vector

$$X = \begin{bmatrix} \text{Traffic} \\ \text{Speed} \\ \text{Rain} \\ \text{Temp} \\ \text{AQI} \end{bmatrix}$$

15.2 Mean Example

$$\mu = \begin{bmatrix} 120 \\ 35 \\ 2 \\ 32 \\ 180 \end{bmatrix}$$

15.3 Covariance Interpretation

- Rain $\uparrow \Rightarrow$ Traffic $\uparrow$
- Rain $\uparrow \Rightarrow$ Speed $\downarrow$
- Traffic $\uparrow \Rightarrow$ Pollution $\uparrow$

Insight: Covariance = dependency map.

15.4 Gaussian Assumption

$$X \sim \mathcal{N}(\mu, \Sigma)$$

Why Gaussian?

- Central Limit Theorem
- Sensor noise
- Analytical simplicity

15.5 Transformation (Congestion Index)

$$Y = 0.6X_1 - 0.4X_2 + 0.8X_3$$

$$Y = a^T X$$

Results:

- Y is Gaussian
- $\mathbb{E}[Y] = a^T \mu$
- $\text{Var}(Y) = a^T \Sigma a$

15.6 Inference

If rainfall increases:

- Speed decreases
- Traffic increases
- Pollution increases

Final Insight:

- Random vector = system state
- Covariance = relationships
- Gaussian = structured uncertainty
- Transformation = derived metrics

Final Summary

- Random vectors simplify multi-variable analysis
- Mean shifts with transformation, covariance does not depend on shift
- Linear transformations preserve Gaussianity
- Covariance matrix is central to understanding dependencies
- Real-world systems (like Smart Cities) can be modeled using these concepts

Source: Lecture 20 Slides :contentReference[oaicite:0]index=0